

CD47 (Cluster of Differentiation 47).

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CD47, also known as integrin-associated protein, is a constitutively and ubiquitously expressed transmembrane receptor. CD47 is conserved across amniotes including mammals, reptiles, and birds. Expression is increased in many cancers and, in non-malignant cells, by stress and with aging. The up-regulation of CD47 expression is generally epigenetic, whereas gene amplification occurs with low frequency in some cancers. CD47 is a high affinity signaling receptor for the secreted protein thrombospondin-1 (THBS1) and the counter-receptor for signal regulatory protein- α (SIRPA, SIRP α) and SIRP γ (SIRPG). CD47 interaction with SIRP α serves as a marker of self to innate immune cells and thereby protects cancer cells from phagocytic clearance. Consequently, higher CD47 correlates with a poor prognosis in some cancers, and therapeutic blockade can suppress tumor growth by enhancing innate antitumor immunity. CD47 expressed on cytotoxic T cells, dendritic cells, and NK cells mediates inhibitory THBS1 signaling that further limits antitumor immunity. CD47 laterally associates with several integrins and thereby regulates cell adhesion and migration. CD47 has additional lateral binding partners in specific cell types, and ligation of CD47 in some cases modulates their function. THBS1-CD47 signaling in non-malignant cells inhibits nitric oxide/cGMP, calcium, and VEGF signaling, mitochondrial homeostasis, stem cell maintenance, protective autophagy, and DNA damage response, and promotes NADPH oxidase activity. CD47 signaling is a physiological regulator of platelet activation, angiogenesis and blood flow. THBS1/CD47 signaling is frequently dysregulated in chronic diseases.

Therapeutic opportunities for targeting the ubiquitous cell surface receptor CD47.

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CD47 is a ubiquitously expressed cell surface receptor that serves as a counter-receptor for SIRP α in recognition of self by the innate immune system. Independently, CD47 also functions as an important signaling receptor for regulating cell responses to stress. We review the expression, molecular interactions, and pathophysiological functions of CD47 in the cardiovascular and immune systems. CD47 was first identified as a potential tumor marker, and we examine recent evidence that its dysregulation contributes to cancer progression and evasion of anti-tumor immunity. We further discuss therapeutic strategies for enhancing or inhibiting CD47 signaling and applications of such agents in preclinical models of ischemia and ischemia/reperfusion injuries, organ transplantation, pulmonary hypertension, radioprotection, and cancer. Ongoing studies are revealing a central role of CD47 for conveying signals from the extracellular microenvironment that limit cell and tissue survival upon exposure to various types of stress. Based on this key function, therapeutics targeting CD47 or its ligands thrombospondin-1 and SIRP α could have broad applications spanning reconstructive surgery, engineering of tissues and biocompatible surfaces, vascular diseases, diabetes, organ transplantation, radiation injuries, inflammatory diseases, and cancer.

Loss of CD47 alters CD8+ T cell activation in vitro and immunodynamics in mice.

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CD47 has established roles in the immune system for regulating macrophage phagocytosis and

lymphocyte activation, with growing evidence of its cell-intrinsic regulatory roles in natural killer and CD8+ T cells. CD47 limits antigen-dependent cytotoxic activities of human and murine CD8+ T cells, but its role in T cell activation kinetics remains unclear. Using in vitro and in vivo models, we show here that CD47 differentially regulates CD8+ T cell responses to short- versus long-term activation. Although CD47 was not required for T cell development in mice and early activation in vitro, short-term stimuli elevated pathogen-reactive gene expression and enhanced proliferation and the effector phenotypes of Cd47-deficient relative to Cd47-sufficient CD8+ T cells. In contrast, persistent TCR stimulation limited the effector phenotypes of Cd47^{-/-} CD8+ T cells and enhanced their apoptosis signature. CD8+ T cell expansion and activation in vivo induced by acute lymphocytic choriomeningitis virus (LCMV) infection did not differ in the absence of CD47. However, the frequency and effector phenotypes of Cd47^{-/-} CD8+ T cells were constrained in chronic LCMV-infected as well as in mice bearing B16 melanoma tumors. Therefore, CD47 regulates CD8+ T cell activation, proliferation, and fitness in a context-dependent manner.

The role of CD47 in non-neoplastic diseases.

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CD47 is a 50 kDa five-spanning membrane receptor that plays a crucial role in multiple cellular processes, including myeloid cell activation, neutrophils transmigration, vascular remodeling, leukocyte adhesion and trans-endothelial migration. Recent studies have revealed that CD47 is a highly expressed anti-phagocytic signal in several types of cancer, and therefore, blocking of CD47 has shown an effective therapeutic potential in cancer immunotherapy. In addition, CD47 has been found to be involved in a complex interplay with microglia and other types of cells, and increasing evidence indicates that CD47 can be targeted as part of immune modulatory strategies for non-neoplastic diseases as well. In this review, we focus on CD47 and its role in non-neoplastic diseases, including neurological disorders, atherosclerosis and autoimmune diseases. In addition, we discuss the major challenges and potential remedies associated with CD47-SIRP α -based immunotherapies.

CD47-SIRP α Axis as a Biomarker and Therapeutic Target in Cancer: Current Perspectives and Future Challenges in Nonsmall Cell Lung Cancer.

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CD47 is a cell surface protein in the immunoglobulin superfamily which is normally expressed at low levels in every healthy cell. It's main physiologic function is to act as an inhibitor of phagocytosis; this occurs throughout interaction with SIRP α expressed on macrophages. Interaction between CD47 and SIRP α leads to activation of tyrosine phosphatases that inhibit myosin accumulation at the submembrane assembly site of the phagocytic synapse, resulting in phagocytosis blockade. In this way CD47 acts as a "don't eat me signal" for healthy self-cells; accordingly, loss of CD47 leads to phagocytosis of aged or damaged cells. Taking advantage of this anti-phagocytic signal provided by CD47, many types of tumors overexpress this protein, thereby avoiding phagocytosis by macrophages and aiding in the survival of cancer cells. The aim of this review is to describe the physiologic the pathophysiologic role of CD47; summarize the available high-quality information about this molecule as a potential biomarker and/or therapeutic target in cancer; finally, we present an in-depth analysis of the available information about

CD47 in association with nonsmall cell lung cancer, EGFR mutations, and tumor microenvironment.

Checkpoint CD47 Function On Tumor Metastasis And Immune Therapy.

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The success of cancer immunotherapy on recognition checkpoints for killing cancer cells has raised a great interest of scientists in understanding new and old methods of immunotherapeutic. CD47 (cluster of differentiation 47) is a cell surface glycoprotein and widely expressed on cells, which belongs to the immunoglobulin (Ig) superfamily as a cell membrane receptor which serves in immune therapy. CD47 is an inhibitory receptor expressed on tumor cell surface and interacts with signal receptor protein- α (SIRP- α , also named CD172a or SHPS-1) which may escape from immune cells such as macrophage and T cells. Meanwhile, tumor cells express high CD47 protein which may secrete exosomes with high CD47 expression. The high CD47 expression-exosomes could serve the tumor metastasis process and provide transfer convenience for tumors on the microenvironment. CD47 on cancer cells can also affect the migration and invasion of cells. The high CD47 expression on tumor or CTC (circulating tumor cell) surface means the stronger migration and invasion and makes them escape from immune cells for phagocytosis such as T cells, NK (natural killer) cells and macrophage, which could be used for diagnosis and prognosis on cancer patients. Meanwhile, targeting CD47 combined with other biomarkers such as EpCAM (epithelial cell adhesion molecule), CD44, etc on cancer surface could be used to isolate CTCs from patients' blood. In terms of treatment, anti-CD47 antibody combined with another antibody such as anti-PD-L1 (programmed death-ligand 1) antibody or drugs such as rituximab, DOX or oxaliplatin also has better therapeutic effects and antitumor function to tumors. Using nanomaterials as an intermediary for CD47-related immune therapy could greatly increase the therapeutic effect and overcome multiple biological barriers for anti-CD47 antibody in vivo. In this review, we discuss the important role and the function of CD47 in tumor metastasis and also provide a reference for related research.

Is CD47 an innate immune checkpoint for tumor evasion?

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Cluster of differentiation 47 (CD47) (also known as integrin-associated protein) is a ubiquitously expressed glycoprotein of the immunoglobulin superfamily that plays a critical role in self-recognition. Various solid and hematologic cancers exploit CD47 expression in order to evade immunological eradication, and its overexpression is clinically correlated with poor prognoses. One essential mechanism behind CD47-mediated immune evasion is that it can interact with signal regulatory protein- α (SIRP α) expressed on myeloid cells, causing phosphorylation of the SIRP α cytoplasmic immunoreceptor tyrosine-based inhibition motifs and recruitment of Src homology 2 domain-containing tyrosine phosphatases to ultimately result in delivering an anti-phagocytic-"don't eat me"-signal. Given its essential role as a negative checkpoint for innate immunity and subsequent adaptive immunity, CD47-SIRP α axis has been explored as a new target for cancer immunotherapy and its disruption has demonstrated great therapeutic promise. Indeed, CD47 blocking antibodies have been found to decrease primary tumor size and/or metastasis in various pre-clinical models. In this review, we highlight the various functions of CD47, discuss anti-tumor responses generated by both the innate and adaptive immune systems as a consequence of administering anti-CD47 blocking antibody, and finally elaborate on the clinical potential of CD47 blockade. We argue that CD47 is a checkpoint molecule for both innate

and adaptive immunity for tumor evasion and is thus a promising target for cancer immunotherapy.

CD47 as a promising therapeutic target in oncology.

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CD47 is ubiquitously expressed on the surface of cells and plays a critical role in self-recognition. By interacting with SIRP α , TSP-1 and integrins, CD47 modulates cellular phagocytosis by macrophages, determines life span of individual erythrocytes, regulates activation of immune cells, and manipulates synaptic pruning during neuronal development. As such, CD47 has recently be regarded as one of novel innate checkpoint receptor targets for cancer immunotherapy. In this review, we will discuss increasing awareness about the diverse functions of CD47 and its role in immune system homeostasis. Then, we will discuss its potential therapeutic roles against cancer and outlines, the possible future research directions of CD47- based therapeutics against cancer.

[Latest Findings on the Role of CD47 in Tumor Immune Evasion and Related Targeted Therapies].

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CD47 is an immunoglobulin that is overexpressed on the surface of a variety of cancer cells. CD47 forms a signaling complex with signal regulatory protein alpha (SIRP α), prompting the escape of cancer cells from macrophage-mediated phagocytosis. In recent years, CD47 has been shown to be highly expressed in many types of solid tumors and is associated with poor prognosis in patients. More and more studies have shown that inhibition of the CD47-SIRP α signaling pathway can promote adaptive immune responses and enhance the phagocytosis of tumor cells by macrophages. Humanized anti-CD47 IgG4 monoclonal antibody has been studied in clinical trials for the treatment of a variety of advanced solid tumors and lymphomas, demonstrating a sound safety profile and achieving partial remission in some patients. In this review we discuss the structure and function of CD47 and the mechanism of CD47 regulation in tumors, summarize the research progress in therapeutic antibody drugs targeting CD47 and a bottleneck in research that targeted drugs are more prone to result in serious adverse effects, and evaluated the potential of the applying CD47-SIRP α signaling pathway in anti-cancer therapy.

Smooth muscle cell-specific CD47 deletion suppresses atherosclerosis.

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Recent smooth muscle cell (SMC)-lineage tracing and single-cell RNA sequencing (scRNA-seq) experiments revealed a significant role of SMC-derived cells in atherosclerosis development. Further, thrombospondin-1 (TSP1), a matricellular protein, and activation of its receptor cluster of differentiation (CD) 47 have been linked with atherosclerosis. However, the role of vascular SMC TSP1-CD47 signaling in regulating VSMC phenotype and atherogenesis remains unknown. We investigated the role of SMC CD47 activation by TSP1 in regulating VSMC phenotype and atherosclerosis development using various in vitro cell-based assays, molecular biological techniques, immunohistological approaches, reanalysis of publicly

available scRNA-seq data, and cell-specific knockout mice. We observed elevated TSP1 expression in human atherosclerotic vascular tissues and VSMCs. TSP1-treated VSMCs exhibited decreased expression of contractile SMC markers (ACTA2, CNN1, and TAGLN) and increased proliferation. Additional experiments and reanalysis of the scRNA-seq dataset showed CD47 as the major TSP1 receptor in VSMCs, with its expression increased in SMC-derived modulated cells of murine atherosclerotic arteries. Knockdown of CD47 gene in human VSMCs upregulated expression of contractile SMC markers and abrogated TSP1's effects on these genes. SMC-specific Cd47 deletion in mice suppressed atherosclerotic lesion formation, reduced macrophage accumulation, and decreased necrotic area. However, no significant differences were observed in weight gain, liver and adipose tissue mass, plasma total cholesterol, and fasting blood glucose between control and SMC-restricted Cd47-deficient mice. Further experiments demonstrated increased efferocytosis of apoptotic CD47-silenced VSMCs by macrophages. These findings suggest that CD47 plays a crucial role in regulating VSMC phenotype, and SMC-specific-Cd47 deletion suppresses atherosclerosis. VSMC phenotypic switching contributes to atherosclerosis development. The present study reports the novel observations that Cd47 levels are upregulated in phenotypically modulated SMCs within atherosclerotic arteries and targeted deletion of Cd47 specifically in SMCs attenuates atherosclerosis. Mechanistic in vitro investigations further showed that TSP1-CD47 signaling regulates VSMC phenotype. Therefore, targeting SMC CD47 represents a promising therapeutic target to suppress atherogenesis.
